

Conjecture 00000000116: A Direct Proof and a Nonidentity Witness

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1 Exact statement and definitions

The repository's English statement is:

Conjecture: There exists a bijection π of $\mathbb{N}$ such that for every n , $|\pi(n) - n| \in \mathbb{P} \cup \{0\}$ (every displacement is a prime or zero).

The Chinese statement has the same quantifiers and also permits zero displacement. Neither version excludes the identity or requires infinitely many moved points. The assertion is elementary: the identity already proves it. Below we also give a nonidentity witness; no mathematical novelty is claimed.

We use the usual definition: p is prime if $2 \leq p$ and every positive divisor d of p is either 1 or p . In particular, 2 is prime: if $2 = dk$ with $d > 0$, then $k \neq 0$, so $1 \leq d \leq 2$ and $d = 1$ or $d = 2$. This is the definition used by the Lean predicate `Prime`, not a predicate restricted to the witness.

Lean's natural subtraction is truncated. We therefore define

$$\text{absDiff}(a, b) = (a \dot{-} b) + (b \dot{-} a), \quad a \dot{-} b = \max(a - b, 0).$$

For $a \leq b$ the sum is $b - a$; otherwise it is $a - b$. Thus it is exactly $|a - b|$. The formal theorem `absDiff_piecewise` proves this distinction. One direction of truncated subtraction alone would not suffice.

2 Proof and nonidentity strengthening

Theorem 1. *The original conjecture holds. Moreover, it has a nonidentity involutive solution that moves exactly 1 and 3.*

Proof. For the original assertion alone, the identity $\pi(n) = n$ is a bijection and has displacement zero at every input. This is a complete proof as written.

For the strengthening, set

$$s(n) = \begin{cases} 3, & n = 1, \\ 1, & n = 3, \\ n, & n \notin \{1, 3\}. \end{cases}$$

The three cases show that $s(s(n)) = n$ for every natural number n . If $s(a) = s(b)$, applying s gives $a = b$, proving injectivity. For any target b , its preimage is $s(b)$, proving surjectivity. The displacements at 1 and 3 are $|3 - 1| = |1 - 3| = 2$, which is prime; all other displacements are zero. Finally $s(1) = 3 \neq 1$ and $s(3) = 1 \neq 3$, while every other element is fixed. This proves all claimed properties. $\square$

3 Whether natural numbers include zero

Lean's `Nat` contains zero, and our witness satisfies $s(0) = 0$. For every $n > 0$, also $s(n) > 0$: the moved points have positive images, and every other positive point is fixed. The restriction to $\mathbb{N}_{>0}$ is therefore well defined and remains self-inverse, hence bijective. Its nonidentity property and displacements are unchanged. A separate Lean theorem establishes the existential statement on the subtype `PositiveNat = {n : Nat // 0 < n}`, so both conventions are covered.

4 Formal statement correspondence

All declarations are in namespace `Conjecture00000000116` in `lean4/Main.lean`. The main theorem has exactly the original existential and universal quantifiers, with no extra hypotheses.

Declaration	Mathematical role
<code>Prime, prime_two</code>	Positive-divisor primality; 2 is prime.
<code>absDiff_pieewise</code>	Absolute difference agrees with its usual piecewise definition.
<code>IsBijection</code>	Injectivity and surjectivity on the whole domain.
<code>AllowedDisplacements</code>	Prime-or-zero displacement for every input.
<code>identity_solution</code>	Independent minimal proof using the identity.
<code>original_conjecture</code>	Original statement, witnessed by the transposition.
<code>nonidentity_strengthening</code>	The involution, exchange, fixed-point and nonidentity properties.
<code>positive_naturals_solution</code>	Nonidentity solution when natural numbers start at one.

The proofs are about all natural numbers, not an assumed solution, a finite array, or a hard-coded Boolean check.

5 Reproduction and trust boundary

The project pins `leanprover/lean4:v4.33.1`. It uses only the bundled `Std` library, with no `Mathlib` or external Lake packages. With Python 3.11 or newer and the pinned toolchain installed, run from the submission directory:

```
python reproduce.py --lake /absolute/path/to/lake
```

On Windows the executable may be named `lake.exe`. The script runs finite regression checks, `lake build`, direct checking of `Main.lean`, and `Check.lean`. The checker displays definitions, principal theorem types, and transitive dependency audits for all 17 public theorems. Every one has an empty axiom list in the tested Lean 4.33.1 build. There are no unproved placeholders, new assumptions, or native evaluation shortcuts. The small concrete facts use ordinary kernel-checked proofs. An empty dependency list does not remove Lean’s kernel or implementation from the trusted computing base.

The default Python tests cover inputs $0, \dots, 10000$, a 65×65 absolute-difference grid, small primality examples, and three large integer samples. These are *regression checks only*, not a proof of an infinite statement. The Lean theorems provide that proof. The optional `-regression-only` mode explicitly reports that Lean was not run. Command failures, version mismatches, and failed or nonempty dependency audits cause the full reproduction command to fail. PDF generation and independent review are separate from this script.

Source identification and attribution

Source: `conjectures/00000000116.md` in the repository:
`The-Last-Math-Competition/The-Last-Math-Competition`.

Commit: `95acb520ec5607c826b8a997b1ef2fc82d6f7c57`
Git blob: `ed6fd48c6da15dfe80337f0253593c3f78550c42`
SHA-256 of the Git blob contents, not CRLF checkout bytes:
`d035f3d5701f9b6abf631c46d3e0410c033af6ec32980195c1baa0cf53ba09e6`

Prepared with AI assistance under the account name `idealstichacker`. No affiliation, award, mathematical novelty, human verification, or upstream acceptance is claimed. Independent AI release review is recorded separately. The proof needs no external mathematical result beyond the elementary arithmetic made explicit above.